Trajectory Planning

1. Content Description

In real robot applications, the robot arm often needs to continuously complete multi-stage process actions, such as moving from the initial standby point to the grasp preparation point, performing the grasp, moving to the transition relay point, and finally moving to the target placement point.

This course introduces how to implement continuous planning and execution of multi-segment joint trajectories in MoveIt2 (`multi_track_motion`). The program pre-draws the spatial motion trajectory lines of the robot arm's end effector between multiple stages in RViz through forward kinematics interpolation, and then sequentially schedules the controller to drive the simulated and physical robot arms to smoothly complete the target pose switches of each stage.

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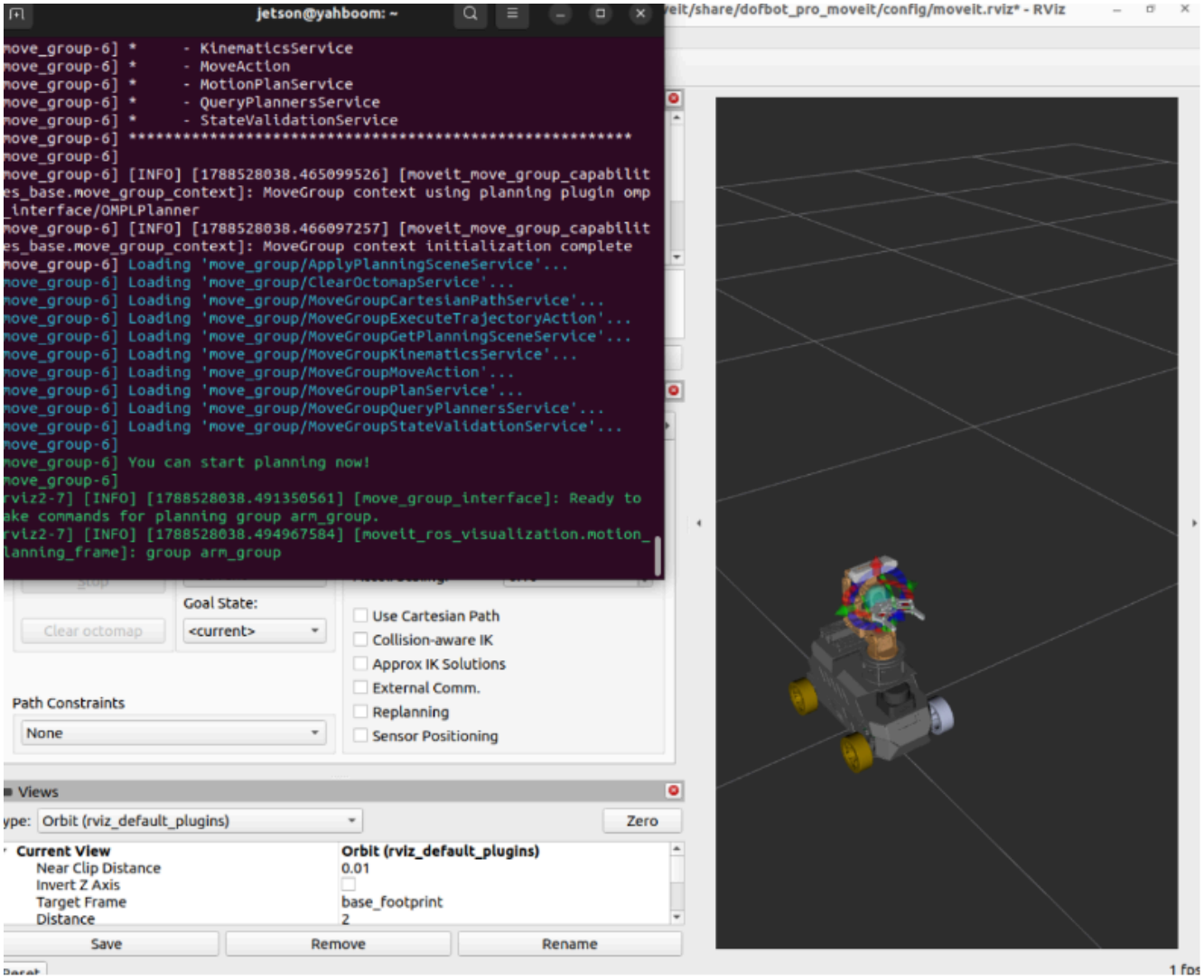
2. Launch and Operation Steps

2.1 Launching the MoveIt2 Simulation Environment

Enter the following command in the terminal to launch the MoveIt2 simulation environment:

```
ros2 launch dofbot_pro_moveit demo_3d.launch.py
```

Once the program starts, when **You can start planning now!** is output in the terminal, it means the MoveIt2 core nodes are fully initialized, as shown in the figure below:



2.2 Starting Real-Machine Linkage (Optional)

If you want the simulated robot arm motion to synchronously drive the physical robot arm, open another terminal and run the real hardware control conversion node:

```
ros2 run moveIt_demo SimulationToMachine
```

(It is recommended not to start this node during the initial test. First verify that the obstacle-avoidance trajectory is correct in the pure RViz simulation environment before connecting to the real machine).

2.3 Adding the Trajectory Visualization Plugin in RViz

In order to observe the green spatial lines planned for each stage trajectory in real time in RViz, the `MarkerArray` display plugin needs to be added:

1. Click the 【Add】 button in the lower left corner of the RViz interface, select 【MarkerArray】 in the list, and click 【OK】 to add it, as shown in the figure below:
2. In the Displays list on the left, expand 【MarkerArray】 and change its subscribed Topic to `/plan_point`, as shown in the figure below:

2.4 Running the Multi-Segment Trajectory Planning Program

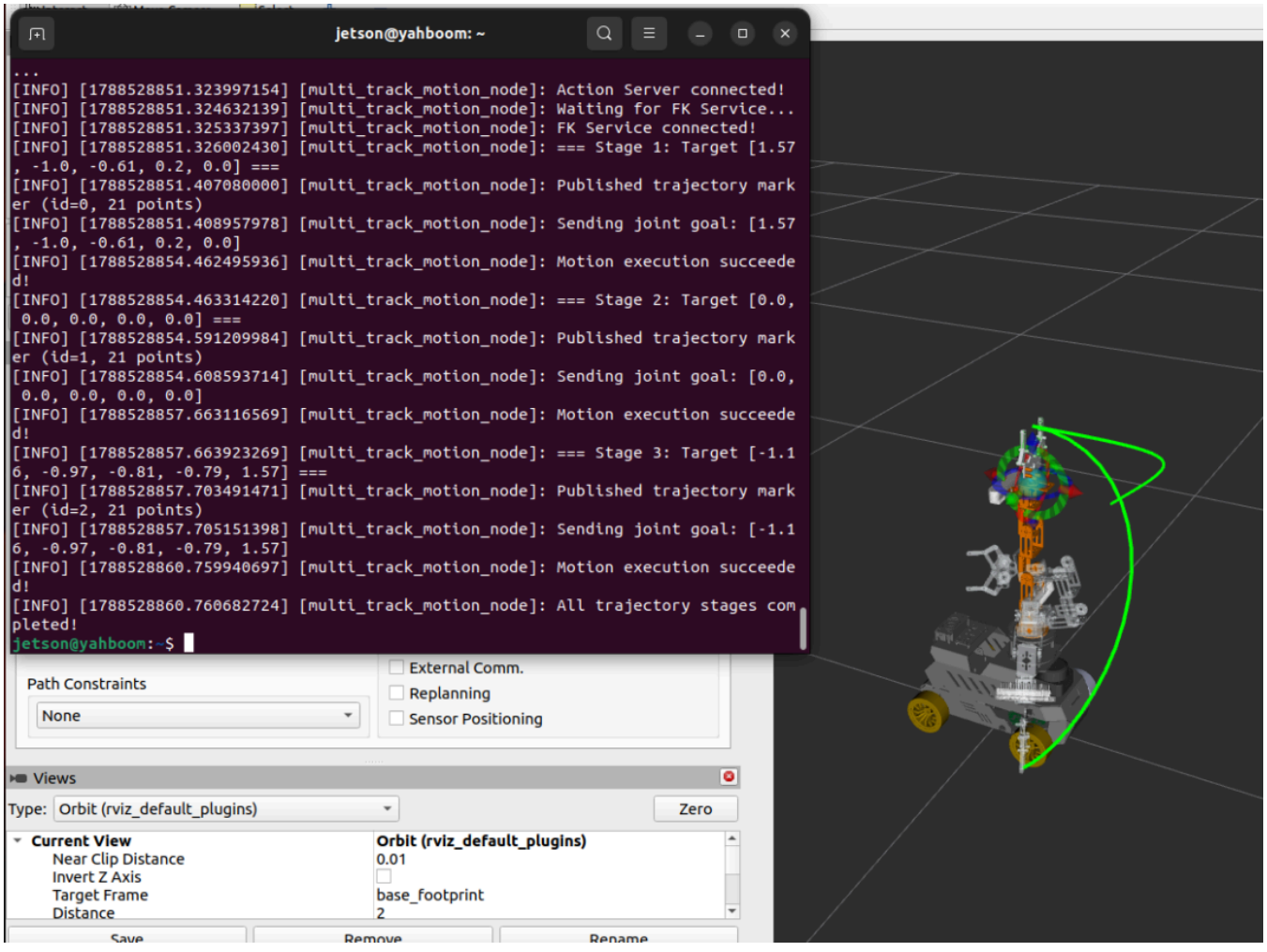
In another terminal, enter the following command to launch the multi-segment trajectory planning node:

```
ros2 run moveIt_demo multi_track_motion
```

After the program starts, the robot arm will sequentially execute the target actions of 3 typical stages:

- **Stage 1**: The robot arm bends and stretches forward to the left, and RViz draws the first green spatial trajectory line;
- **Stage 2**: The robot arm smoothly returns to the vertical origin pose, drawing the second return trajectory;
- **Stage 3**: The robot arm tilts backward to the right and rotates the end effector, drawing the third action trajectory.

The entire multi-segment motion process is coherent and smooth, and RViz will retain and dynamically display the motion trajectory lines of each stage, as shown in the figure below:



3. Core Code Analysis

Program source code path:
`~/yahboomcar_ros2_ws/dofbot_pro_ws/src/movelt_demo/movelt_demo/multi_track_motion.py`

3.1 Node Initialization and Communication Interfaces

The program is written in native ROS 2 Python. The class `MultiTrackMotion` initializes the trajectory execution Action client, the Marker visualization publisher, and the forward kinematics (FK) service client:

```
import math
import rclpy
from rclpy.node import Node
from rclpy.action import ActionClient
from control_msgs.action import FollowJointTrajectory
from trajectory_msgs.msg import JointTrajectory, JointTrajectoryPoint
from builtin_interfaces.msg import Duration
from geometry_msgs.msg import Point
from moveit_msgs.srv import GetPositionFK
from visualization_msgs.msg import Marker, MarkerArray
from std_msgs.msg import ColorRGBA

class MultiTrackMotion(Node):
    def __init__(self):
        super().__init__('multi_track_motion_node')
        self.get_logger().info("Initializing MultiTrackMotion node...")

        # 1. Trajectory execution Action client
        self._action_client = ActionClient(
            self,
            FollowJointTrajectory,
            '/arm_group_controller/follow_joint_trajectory'
        )

        # 2. The 5 controlled joint names
        self.joint_names = [
            "arm1_Joint", "arm2_Joint", "arm3_Joint",
            "arm4_Joint", "arm5_Joint"
        ]

        # 3. MarkerArray trajectory line publisher (topic: /plan_point)
        self._marker_pub = self.create_publisher(MarkerArray, '/plan_point', 10)
        self._marker_id = 0

        # 4. Forward kinematics (FK) computation client (service: /compute_fk)
        self._fk_client = self.create_client(GetPositionFK, '/compute_fk')
```

3.2 Joint-Space Interpolation and End-Effector Forward-Kinematics Trajectory Visualization

Before each segment of motion is executed, in order to present the future spatial walking path of the robot arm's end effector to the operator, the `publish_trajectory_line` function performs equidistant interpolation between the start joint angles and the target joint angles (20 discrete steps by default), calls the `/compute_fk` service to obtain the spatial 3D coordinates (x, y, z) of the end link `Gripping` at each interpolation point, packages them as a `Marker.LINE_STRIP` green line, and publishes them to RViz:

```
    def compute_fk_point(self, joint_positions):
        """Call the /compute_fk service to solve for the end-effector 3D spatial position at
the given joint angles"""
        fk_request = GetPositionFK.Request()
        fk_request.header.frame_id = "base_link"
        fk_request.header.stamp = self.get_clock().now().to_msg()
        fk_request.fk_link_names = ["Gripping"]
        fk_request.robot_state.joint_state.name = self.joint_names
        fk_request.robot_state.joint_state.position = list(joint_positions)

        future = self._fk_client.call_async(fk_request)
        rclpy.spin_until_future_complete(self, future)
        response = future.result()

        if response.error_code.val == response.error_code.SUCCESS:
            pos = response.pose_stamped[0].pose.position
            return Point(x=pos.x, y=pos.y, z=pos.z)
        return None

    def publish_trajectory_line(self, start_joints, target_joints, num_points=20):
        """Linearly interpolate in joint space, compute the end-effector Cartesian
trajectory, and publish a LINE_STRIP Marker"""
        cartesian_points = []
        for i in range(num_points + 1):
            t = i / num_points
            # Linear interpolation of joint angles
            interp_joints = [
                start_joints[j] + t * (target_joints[j] - start_joints[j])
                for j in range(len(start_joints))
            ]
            pt = self.compute_fk_point(interp_joints)
            if pt:
                cartesian_points.append(pt)

        # Build the MarkerArray message
        marker_array = MarkerArray()
        marker = Marker()
        marker.header.frame_id = "base_link"
        marker.header.stamp = self.get_clock().now().to_msg()
        marker.ns = "multi_track"
        marker.id = self._marker_id
        self._marker_id += 1
        marker.type = Marker.LINE_STRIP
```

```
        marker.action = Marker.ADD
        marker.scale.x = 0.005  # Line width 5mm
        marker.color = ColorRGBA(r=0.0, g=1.0, b=0.0, a=1.0)  # Bright green
        marker.points = cartesian_points
        marker.lifetime = Duration(sec=60, nanosec=0)  # Keep for 60 seconds for easy
observation
        marker_array.markers.append(marker)

        self._marker_pub.publish(marker_array)
```

3.3 Trajectory Execution and Synchronous Waiting

The target angles of each stage are sent to the trajectory controller by constructing a `FollowJointTrajectory.Goal`, with the motion duration set to 3.0 seconds. A synchronous waiting mechanism is used to ensure that the current stage action is fully executed before entering the next stage:

```
    def move_to_joints(self, joint_positions, move_time=3.0):
        """Send the joint goal and synchronously wait for the execution result"""
        goal_msg = FollowJointTrajectory.Goal()
        trajectory = JointTrajectory()
        trajectory.joint_names = self.joint_names

        point = JointTrajectoryPoint()
        point.positions = joint_positions
        point.velocities = [0.0] * len(joint_positions)
        point.time_from_start = Duration(
            sec=int(move_time),
            nanosec=int((move_time % 1) * 1e9)
        )

        trajectory.points.append(point)
        goal_msg.trajectory = trajectory

        send_goal_future = self._action_client.send_goal_async(goal_msg)
        rclpy.spin_until_future_complete(self, send_goal_future)

        goal_handle = send_goal_future.result()
        if not goal_handle.accepted:
            self.get_logger().error("The target trajectory was rejected by the controller!")
            return False

        result_future = goal_handle.get_result_async()
        rclpy.spin_until_future_complete(self, result_future)
        return result_future.result().result.error_code == 0
```

3.4 Multi-Stage State Machine Scheduling (run Method)

In the main execution logic, stage parameters for 3 different poses are defined, and the trajectories are drawn, sent for execution, and the current joint angles of the state machine are maintained sequentially:

```
    def run(self):
        # Define the joint target radians of the 3 stages
```

```
        motions = [
            {"name": "Stage 1", "joints": [1.57, -1.00, -0.61, 0.20, 0.0]},
            {"name": "Stage 2", "joints": [0.0,   0.0,   0.0,  0.0,  0.0]},
            {"name": "Stage 3", "joints": [-1.16, -0.97, -0.81, -0.79, 1.57]},
        ]

        current_joints = [0.0] * 5

        for motion in motions:
            target = motion["joints"]
            name = motion["name"]
            self.get_logger().info(f"=== Executing {name}: target angles {[round(j, 2) for j
in target]} ===")

            # 1. Pre-compute and visualize the trajectory line
            self.publish_trajectory_line(current_joints, target)

            # 2. Send the motion for execution (motion duration 3.0s)
            success = self.move_to_joints(target, move_time=3.0)
            if not success:
                self.get_logger().error(f"{name} failed to execute, terminating subsequent
actions!")
                return

            current_joints = target[:]

        self.get_logger().info("All multi-segment trajectory planning tasks completed
successfully!")
```